ICT305 Topics in IT

Quantum Computing

CIHE Australia

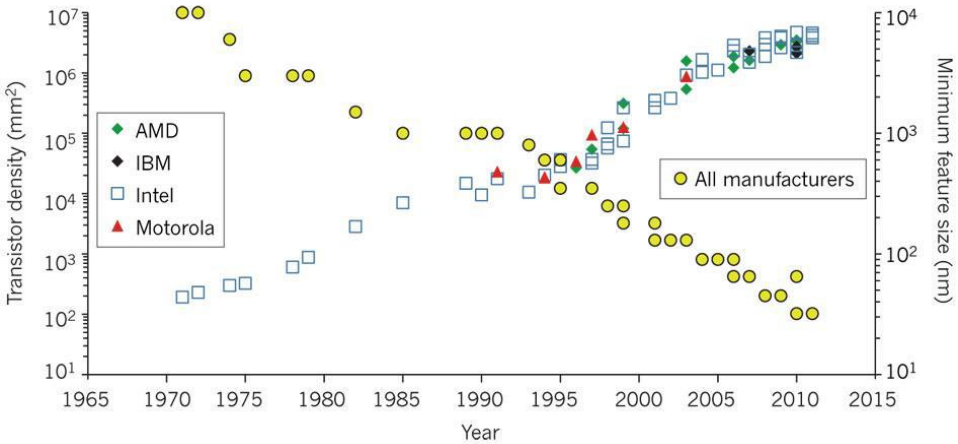
Agenda

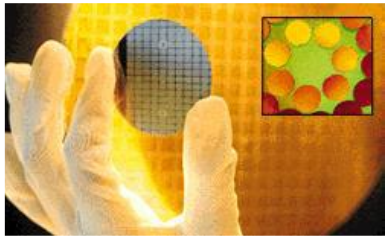
- What is a Quantum Computer?
- Basic Concepts of Quantum Computers
- How does a Quantum Computer work?
- Applications of Quantum Computers

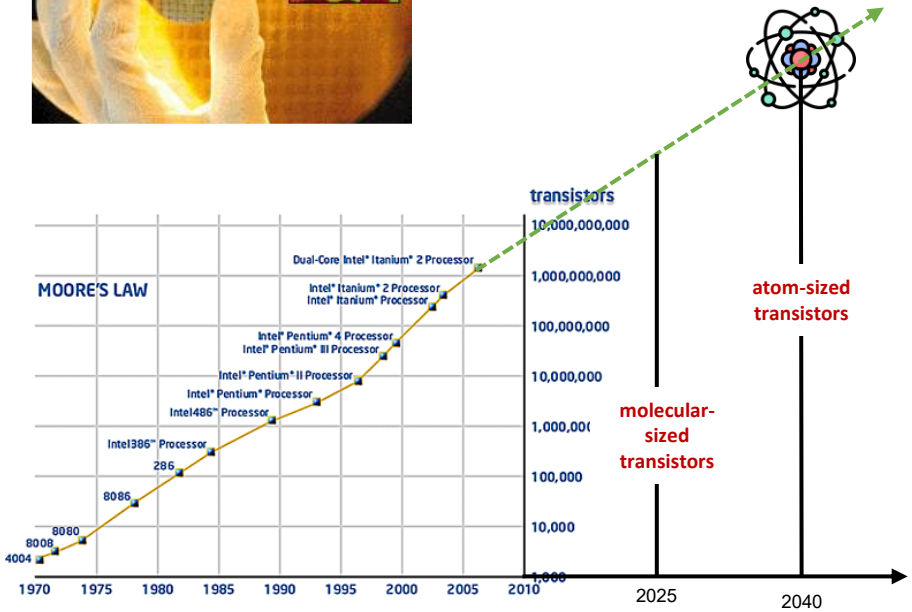
References

1. Shubin Liu, *Introduction to Quantum Computers, Workshop,* University of North Carolina at Chapel Hill

Moore's law & Future of Computers

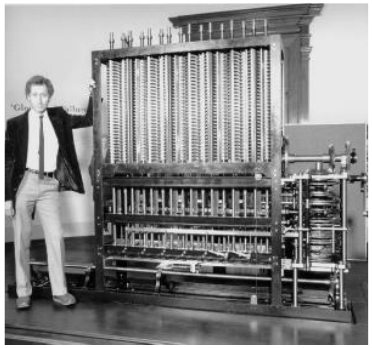




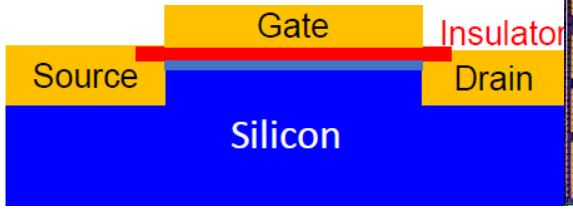


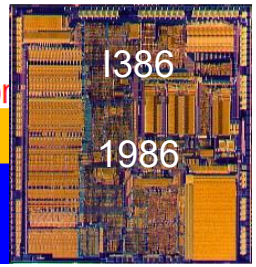
*Moore's law*is the observation that the number of transistors in a dense integrated circuit doubles about every two years.

Moore's law will soon run into major physical constraints!



Moore, G. E. *Electronics* **8**, 114–117 (1965).
Image from: Ferain, I. *et al.*, *Nature* **479**, 310–316 (2011).



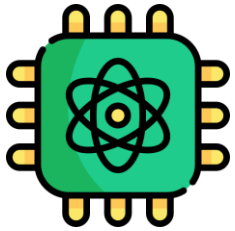
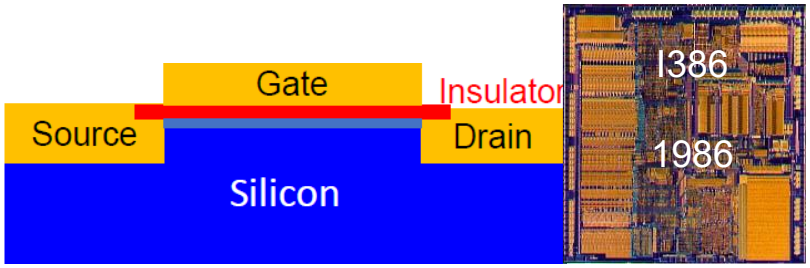


"There's Plenty of Room at the Bottom" (1959)

"When we get to the very, very small world –say circuits of seven atoms –we have a lot of new things that would happen that represent **completely new opportunities for design**. *Atoms on a small scale behave like nothing on a large scale, for they satisfy the laws of quantum mechanics…"*

Richard Feynman

What is a quantum computer?

Quantum Computer	Classical Computer
A computer that **uses laws of quantum mechanics** to perform massively parallel computing through superposition, entanglement, and decoherence. 	A computer that **uses voltages flowing through circuits and gates**, which can be controlled and manipulated entirely by classical mechanics. 

Evolution of Quantum Theory & Quantum Technology

The Foundations

1929: Diract, one of the founder of quantum physics writes that physicists now know all laws necessary to simulate chemical systems

1965: Feynman's theories of quantum-electro-dynamics

1935: The EPR paradox is a thought experiment intended to demonstrate an inherent paradox in the early formulations of quantum theory. It is among the best-known examples of quantum entanglement

Brief History of Quantum Computers

- **1981**:Richard Feynman proposed to use quantum computing to model quantum systems. He also describe theoretical model of quantum computer
- **1985**: David Deutsch described first universal quantum computer
- **1994**: Peter Shor developed the first algorithm for quantum computer (factorization into primes)
- **1995** Schumacher proposed “Quantum bit” or “qubit” as physical resource
- **1996**:LovGrover developed an algorithm for search in unsorted database
- **1998**: the first quantum computers on two qubits, based on NMR(Oxford; IBM, MIT, Stanford)
- **2000**: quantum computer on 7 qubits, based on NMR (Los-Alamos)
- **2001**: 15 = 3 x5 on7-qubit quantum computer by IBM
- **2005-2006**: experiments with photons; quantum dots; fullerenes and nanotubes as "particle traps"

Brief History of Quantum Computers

- **2007**: D-Wave announced the creation of a quantum computer on 16 qubits
- **2012**: D-Wave claimed a quantum computation using 84 qubits
- **2017**: D-Wave Systems Inc. announced the D-Wave 2000Q quantum annealer with 2000 qubits
- **2017**: Microsoft revealed Q Sharp with 32 qubits
- **2018**: Google announced the creation of a 72-qubit quantum chip
- **2019**: Google claimed quantum supremacy with 54 qubits to perform operations in 200 seconds that would take a supercomputer about 10,000 years to complete
- **2019**: IBM revealed 53 qubits
- **2020**: Chinese researchers claimed to have achieved quantum supremacy using a photonic 76-qubit system at 100 trillion times the speed of classical supercomputers
- **2020**: IBM will build 1121-qubit quantum computer in 2023, and 1 million-qubit quantum computer in 2030.

Quantum Mechanics

- Quantum mechanics is the theory that describes the behavior of microscopic systems, such as photons, electrons, atoms, molecules, etc.

Nobody understands quantum mechanics!

"No, you're not going to be able to understand it.... You see, my physics students don't understand it either. That is because I don't understand it. Nobody does. ... The theory of quantum electrodynamics describes Nature as absurd from the point of view of common sense. And it agrees fully with an experiment. So, I hope that you can accept Nature as She is –absurd"

--Richard Feynman

Classical vs. Quantum Mechanics

Classical Mechanics

- It deals with macroscopic particles
- It is based on Newton's laws of motion and Maxwell's electromagnetic wave theory
- Any amount of energy may be emitted or absorbed continuously
- The state of a system is defined exactly by specifying their positions and velocities
- The future state can be predicted with certainty

Quantum Mechanics

- It deals with microscopic particles
- It is based on the Schrödinger equation
- In Planck's postulation, only discrete values of energy are emitted or absorbed –origin of "quantum"
- Because of Heisenberg's uncertainty principle and de Broglie hypothesis dual nature of matter (both particle & wave), the state of a system cannot be specified exactly
- It gives probabilities of finding particles at various locations in space

Quantum Mechanics

- **Quantum states**, represented by Dirac's ket, $|\psi>$, evolve in time according to the Schrödinger equation:

$$d|\psi\rangle/dt = -i\hat{H}(t)|\psi\rangle/\hbar,$$

- which implies that time evolution is described by unitary transformations

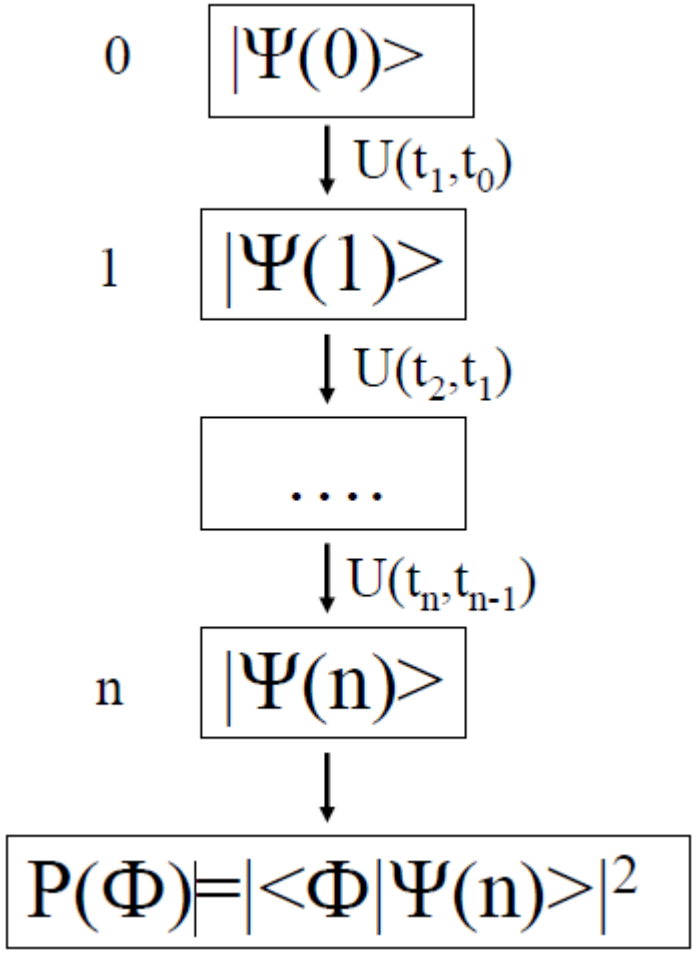
$$|\psi\rangle \rightarrow \hat{U}|\psi\rangle.$$

$$d\hat{U}(t)/dt = -i\hat{H}(t)\hat{U}(t)/\hbar.$$

- where $|\psi>$ is the **quantum state** (wavefunction), and **H** is **Hamiltonian**

- This theory, which has been extensively tested by experiments, is probabilistic in nature.
- The outcomes of measurements on quantum systems are not deterministic.
- Between measurements, quantum systems evolve according to linear equations (the Schrödinger equation).
- This means that solutions to the equations obey a superposition principle: *linear combinations of solutions are still solutions*.

Unitary Transformation as Quantum Computing



PREPARATION:

- The initial preparation of the state defines a wave function at time t0=0.

STATE EVOLUTION:

- Evolved by a sequence of unitary operations

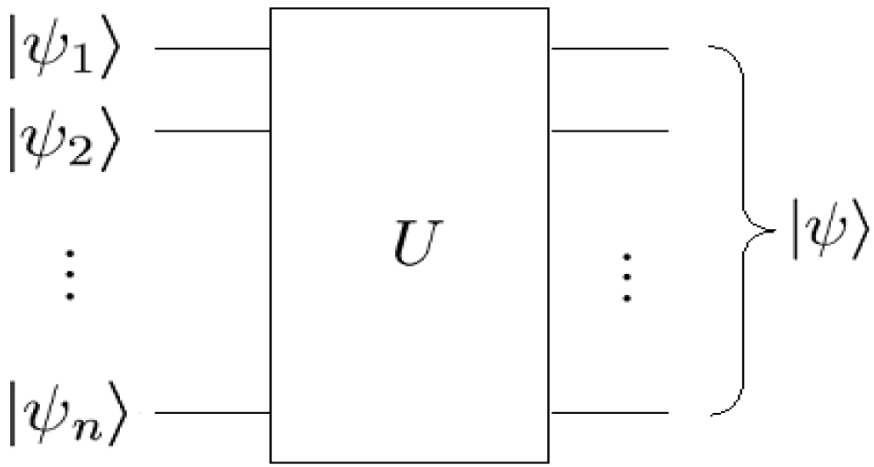
…

MEASUREMENT:

- Quantum measurement is projective.
- Collapsed by measurement of the state

Classical Computing Approach using Flow Chart

Unitary Transformation as Quantum Computing



- On a quantum computer, programs are executed by unitary evolution of an input that is given by the state of the system, $|\psi_n >$, which can in either 0 or 1 state.
- Since all unitary operators are invertible, we can always reverse or 'uncompute' a computation on a quantum computer.

What does a quantum computer look like?

Chinese 76-qubit photon-based quantum computer

IonQ, ion-trap-based 32-qubit quantum computer

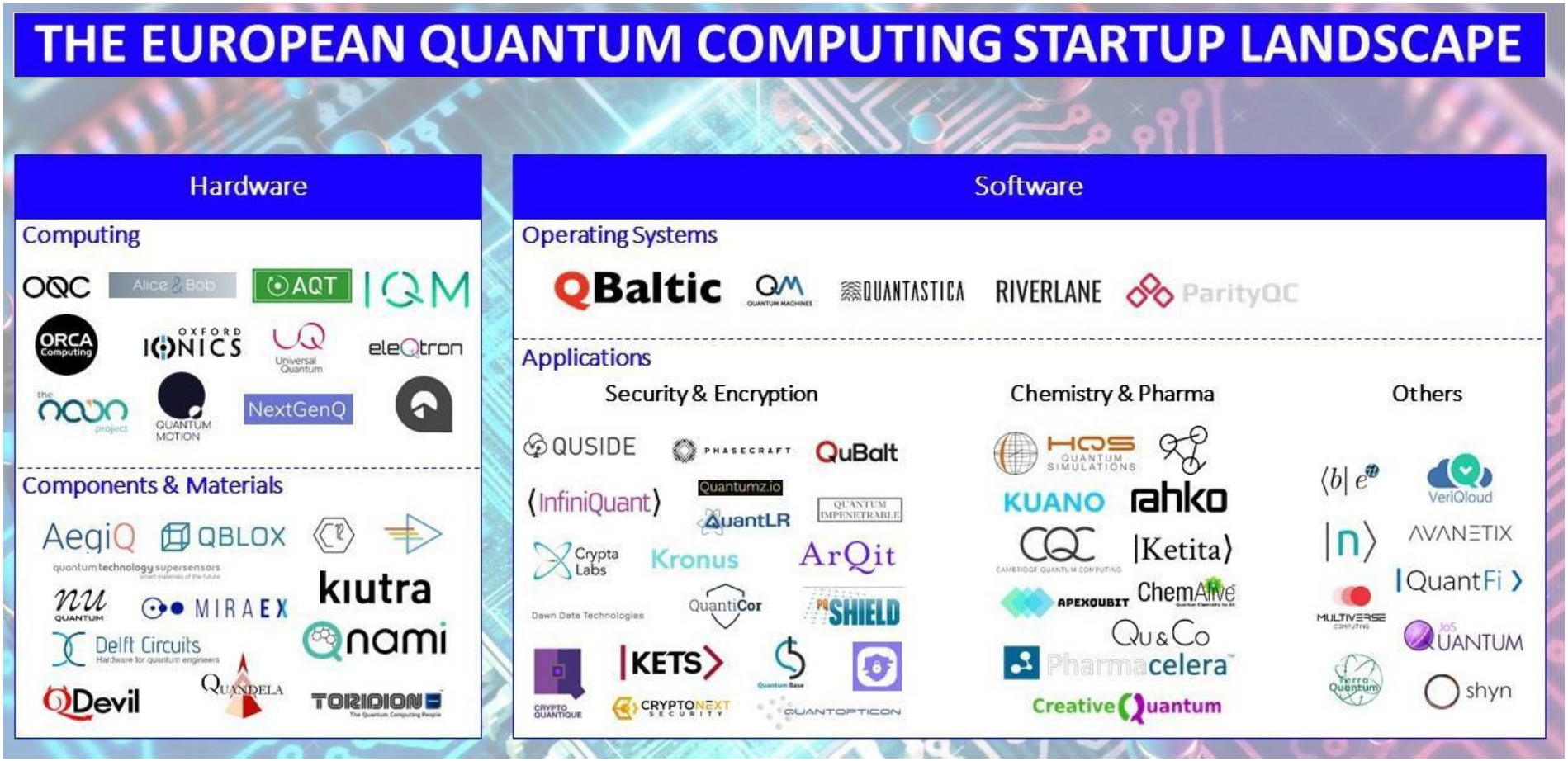
IBM 53-qubit superconductor-based quantum computer

Quantum Computing Race -Corporations

R&D investments reach $10.7 billion by 2024

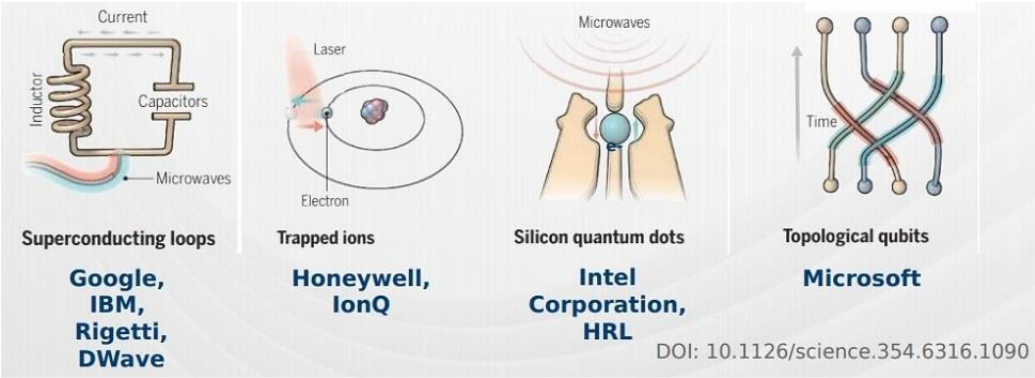
Quantum Computing Race

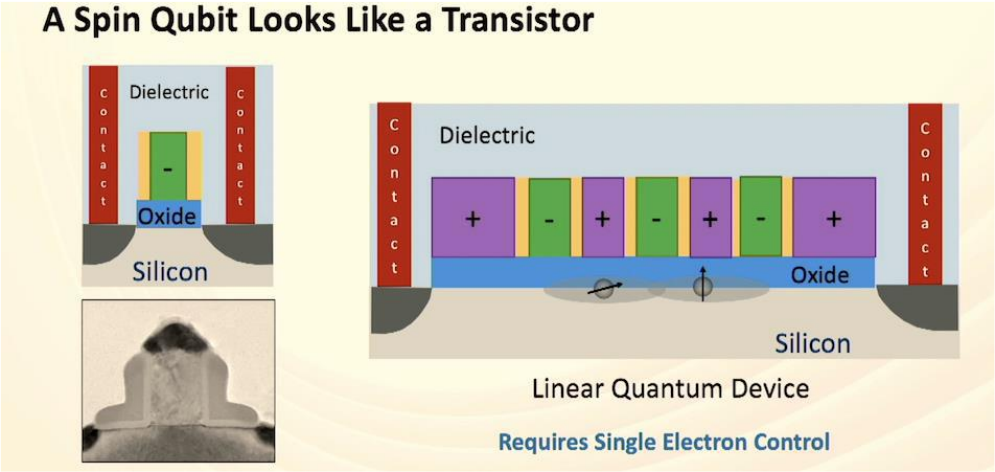
R&D investments reach $10.7 billion by 2024



Quantum Bit -Qubit

- ***The smallest unit of information*** in a quantum computer.
- It represents **the state** of the wavefunction $|\psi_n>$, in Schrödinger equation.
- A **qubit** may be in the "on" (1) state or in the "off" (0) state.
- Many ways to implement a qubit:
  - Nuclear spin in NMR: ↑= |0>, ↓= |1>.
  - Photons in a cavity: 0 photon = |0>, 1 photon = |1>
  - Energy states of an atom: ground state|0>, excited state |1>
  - Polarization of photon, many others….



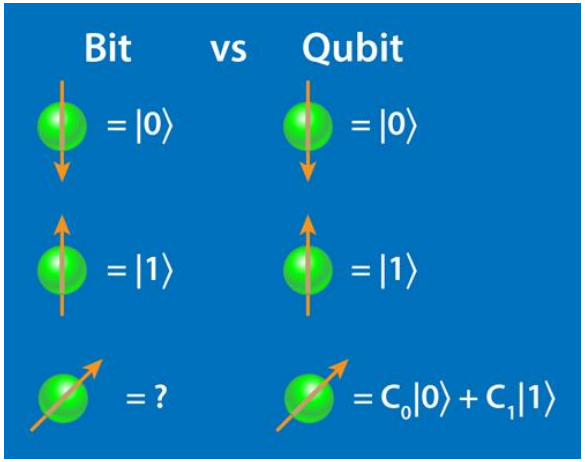


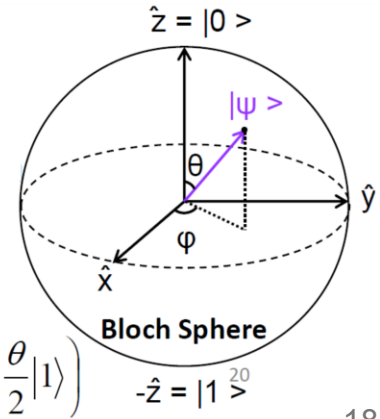
Quantum Bit - Qubit

- Since quantum systems evolve according to linear equations (the Schrödinger equation), linear combinations of solutions are also solutions.
- So, for the state of a qubit $|0\rangle$ and $|1\rangle$, its superposition also describes the same state.
- The general form of a qubit state can be represented by:

$$\alpha_0|0\rangle + \alpha_1|1\rangle,$$

  where α_0 and α_1 are complex numbers that specify the probability amplitudes of the corresponding states.
- $|\alpha_0|^2$ gives the probability that you will find the qubit in the "off" (0) state; $|\alpha_1|^2$ gives the probability that you will find the qubit in the "on" (1) state.
- Normalization condition: $|\alpha_0|^2 + |\alpha_1|^2 = 1$





$$|\Psi\rangle = e^{i\gamma}\left(\cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle\right)$$

Classical Bit vs. Quantum Bit

CLASSICAL BITS:

- can be in two distinct states, 0 and 1
- can be measured completely
- are not changed by measurement
- can be copied
- can be erased

QUANTUM BITS:

- can be in state |0> or in state |1> or in any other state that is a linear combination of the two states
- can be measured partially with given probability
- are changed by measurement
- cannot be copied
- cannot be erased

Advantages of Qubits & Enormous Quantum Power

Adding qubits increases storage exponentially

- Quantum computer doubles the power with every added qubit
  - To double the power of a digital computer 32bits -> 64 bits
  - To double the power of a quantum computer 32qubits -> 33 qubits

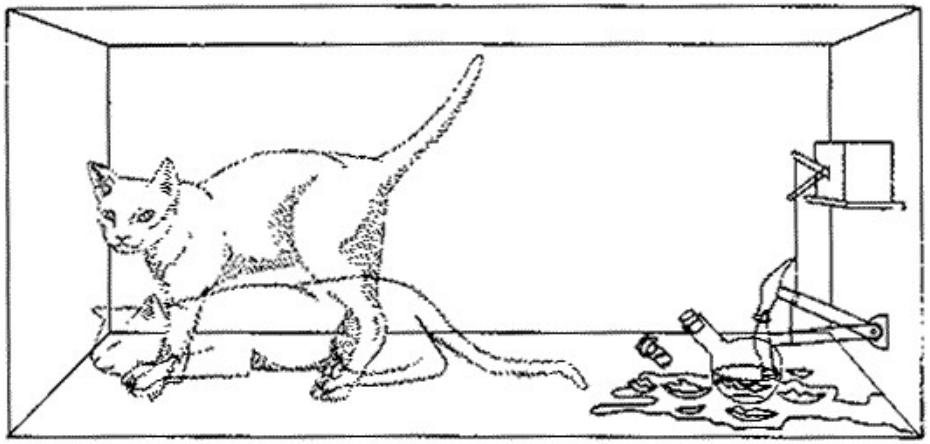
Can do operations on all superpositions…like massively parallel computation

- One math operation on 2^n numbers encoded in classical computers with n bits requires 2^n steps or parallel processors, but the same operation on 2^n numbers encoded by n qubits takes 1 step
  - A 64-bit computer can perform manipulation on 64-bit binary numbers at a time.
  - A 64-qubit quantum computer operates in a space of 2^{64} dimensions, or roughly 16,000,000,000,000,000,000 ($16*10^{18}$) numbers to specify the state of the quantum system.

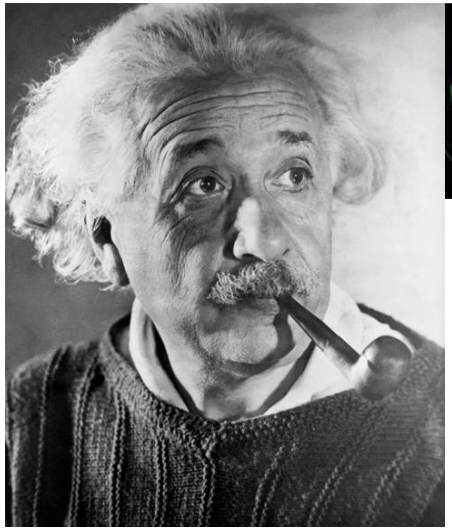
This makes complex problems much easier to solve by quantum computer

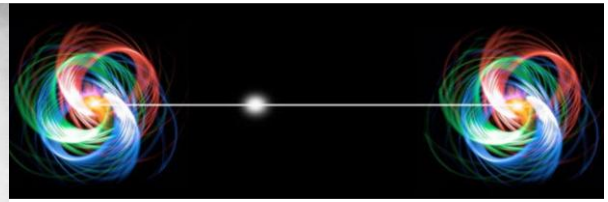
However, Quantum Systems are SPOOKY!

Schrödinger's Cat: Superposition

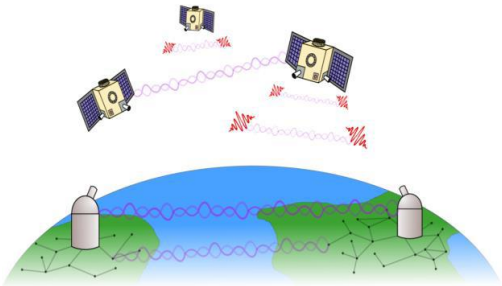


Schrödinger's Cat in a black box





"I cannot seriously believe in [the quantum theory] because it cannot be reconciled with the idea that physics should represent a reality in time and space, free from spooky actions at a distance." *Albert Einstein, March 1947.*



Superposition

- **Every quantum state can be represented as a sum of two or more other distinct states**. Mathematically, it refers to a property of solutions to the Schrödinger equation.
- Since the Schrödinger equation is linear, **any linear combination of solutions** will also be a solution.
- **A single qubit can be forced into a superposition** of the two states denoted by the addition of the state vectors:

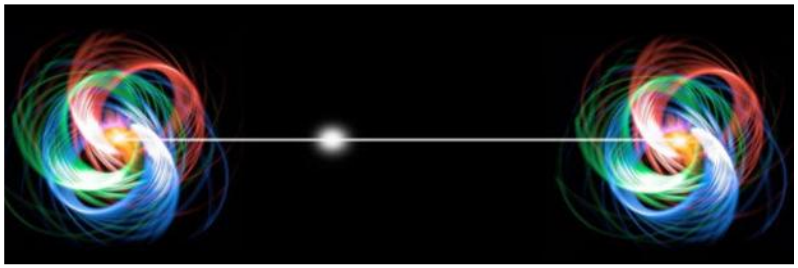
$$|\psi\rangle = \alpha_0\ |0\rangle + \alpha_1|1\rangle, \text{where } |\alpha_0|^2 + |\alpha_1|^2 = 1$$

- **A qubit in superposition** is in both of the states |1> and |0 at the same time
- **If this state is measured**, we see only one or the other state (live or dead) with some probability.
- Consider a 3-qubit register. An equally weighted superposition of all possible states would be denoted by:

$$|\psi\rangle = \frac{1}{\sqrt{8}}|000\rangle + \frac{1}{\sqrt{8}}|001\rangle + \ldots + \frac{1}{\sqrt{8}}|111\rangle$$

Entanglement

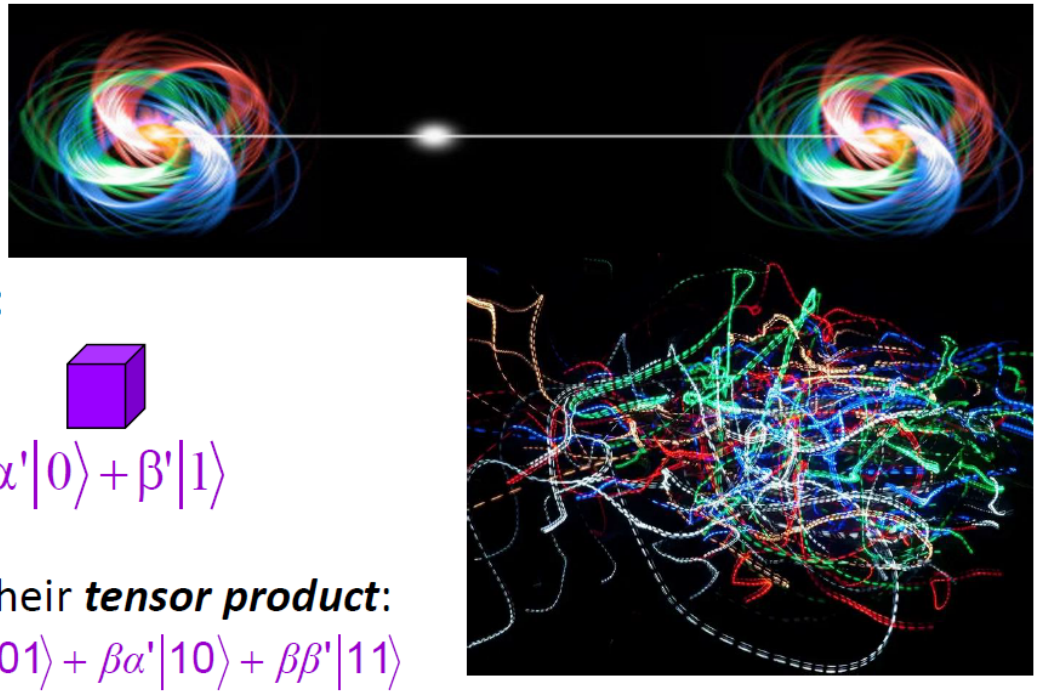
- **Quantum entanglement** is the phenomenon where the quantum state of each particle in a group cannot be described independently of the state of the others, even when the particles are separated by a large distance.



- **An entangled pair** is *a single quantum system* in a superposition of equally possible states. The entangled state contains no information about the individual particles, only that they are in opposite states.
- **If the state of one is changed**, the state of the other is **instantly** adjusted to be consistent with quantum mechanical rules.
- If a measurement is made on one, the other will automatically collapse.
- Entanglement is a primary feature of quantum mechanics lacking in classical mechanics .
- Entanglement is a joint characteristic of two or more quantum particles.
- Einstein called it "spooky actions at a distance"

Entanglement

Entanglement



Suppose that two qubits are in states:

$$\alpha|0\rangle + \beta|1\rangle$$

$$\alpha'|0\rangle + \beta'|1\rangle$$

The state of the combined system is their ***tensor product***:

$$(\alpha|0\rangle + \beta|1\rangle)(\alpha'|0\rangle + \beta'|1\rangle) = \alpha\alpha'|00\rangle + \alpha\beta'|01\rangle + \beta\alpha'|10\rangle + \beta\beta'|11\rangle$$

Question: what are the states of the individual qubits for

1. $\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$ an ***independent*** state
2. $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$ an ***entangled*** state

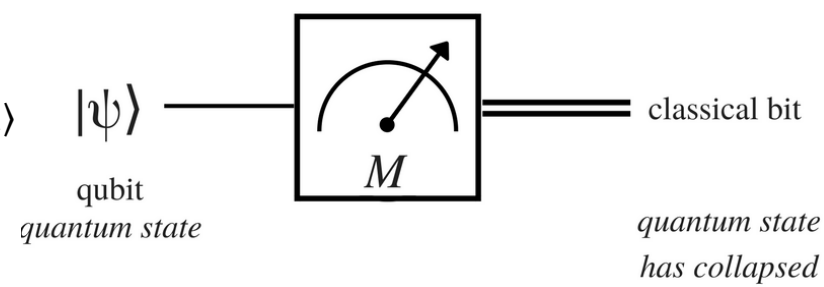
Measurement

- **If a quantum system were perfectly isolated**, it would maintain coherence indefinitely, but it would be impossible to manipulate or investigate it.
- **A quantum measure** is a decoherence process.
- **When a quantum system is measured**, the wave function $|\psi\rangle$ collapses to a new state according to a probabilistic rule.
- If $|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$, after measurement, either $|\psi\rangle = |0\rangle$ or $|\psi\rangle = |1\rangle$ and these alternatives occur with certain probabilities of $|\alpha_0|^2$ and $|\alpha_1|^2$ with $|\alpha_0|^2 + |\alpha_1|^2 = 1$
- A quantum measurement never produces $|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$

Example: Two qubits:

$$|\psi\rangle = 0.316|00\rangle + 0.447|01\rangle + 0.548|00\rangle + 0.632|11\rangle$$

The probability to read the rightmost bit as 0 is $|0.316|^2 + |0.548|^2 = 0.4$



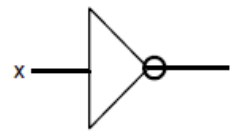
Quantum Gate

- A **quantum gate** (or **quantum logic gate**) is a basic quantum circuit operating on qubits.
- They are the building blocks of quantum circuits, like classical logic gates are for conventional digital circuits
- Due to the normalization condition every gate operation U has to be unitary: $UU^* = I$
- The number of qubits in the input and output of the gate must be equal; a gate which acts on n qubits is represented by 2^n x 2^n unitary matrix
- Unlike many classical logic gates, quantum gates are reversible.

Quantum Gate

Operator	Gate(s)		Matrix
Pauli-X (X)	X	NOT	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)	Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)	Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)	H	$H^{-1} = H$ or $H^2 = I$	$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)	S		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)	T		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		XOR	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)	Z		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)			$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

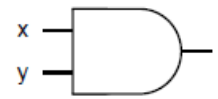
NOT gate



y = NOT(x)

x	y
0	1
1	0

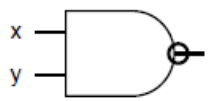
AND gate



z = (x) AND (y)

x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

NAND gate



z = (x) NAND (y)

x	y	z
0	0	1
0	1	1
1	0	1
1	1	0

OR gate

z = (x) OR (y)

x	y	z
0	0	0
0	1	1
1	0	1
1	1	1

NOR gate

z = (x) NOR (y)

x	y	z
0	0	1
0	1	0
1	0	0
1	1	0

XOR gate

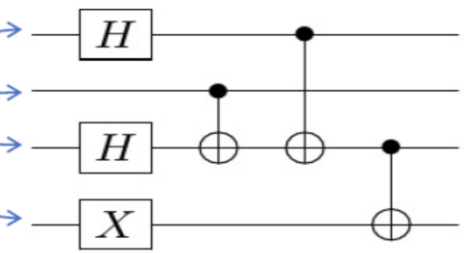
z = (x) XOR (y)

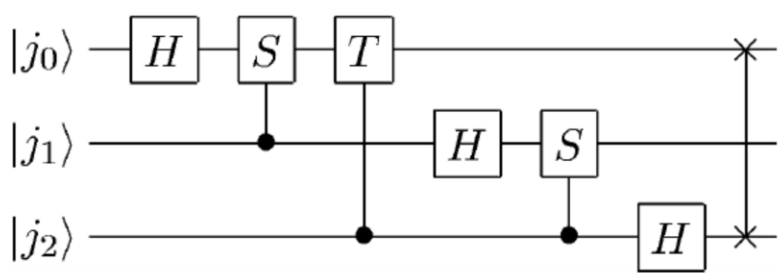
x	y	z
0	0	0
0	1	1
1	0	1
1	1	0

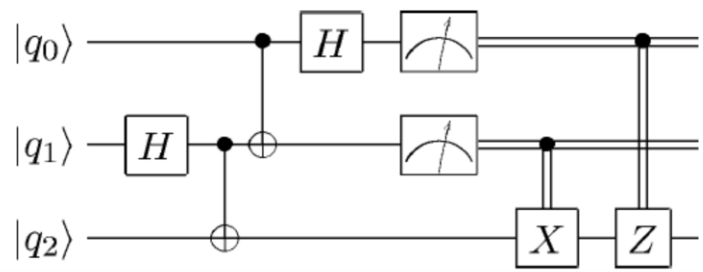
Quantum Circuit

- A quantum circuit is a model for quantum computation in which a computation is a sequence of quantum gates with *n*-qubit register linked by "wires."
- The circuit has fixed "width" corresponding to the number of qubits being processed

Unlike classical circuits, the same number of wires is going throughout the circuit

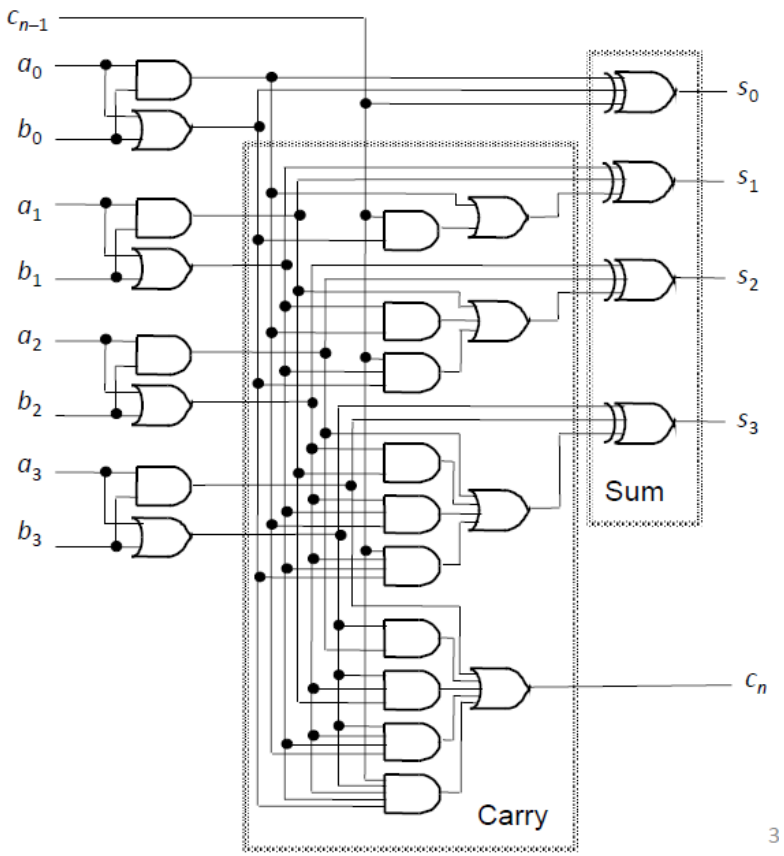




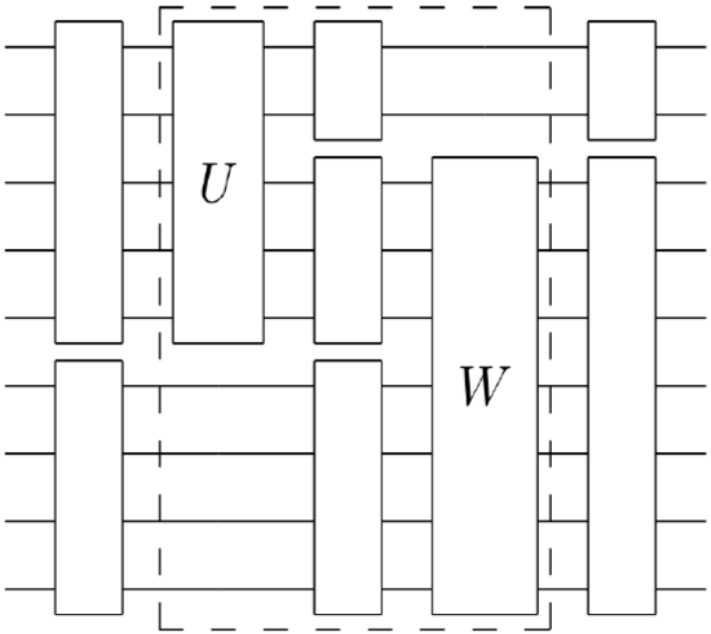


Quantum vs. Classical Circuits

Classical Logic Circuits



Quantum Logic Circuits



Quantum Algorithms

- **It may be possible** to solve a problem on a quantum system much faster (i.e., using fewer steps) than on a classical computer.
- **Factoring and searching** are examples of problems where quantum algorithms are known and are faster than any classical ones.
- Implications for cryptography, information security.
- **What makes a quantum algorithm potentially faster than any classical one?**
  - Quantum parallelism**:** by using superpositions of quantum states, the computer is executing the algorithm on all possible inputs at once.
  - Dimension of quantum Hilbert space**:** the "size" of the state space for the quantum system is exponentially larger than the corresponding classical system.
  - Entanglement capability**:** different subsystems (qubits) in a quantum computer become entangled, exhibiting nonclassical correlations.

Famous Quantum Algorithms

Algorithms	Classical steps	quantum logic steps
Fourier transform e.g.: - Shor's prime factorization - discrete logarithm problem - Deutsch Jozsa algorithm	$N\log(N) = n\,2^n$ $N = 2^n$ - n qubits - N numbers	$\log^2(N) = n^2$ - hidden information! - Wave function collapse prevents us from directly accessing the information
Search Algorithms	N	$\sqrt{N}$
Quantum Simulation	c^N bits	kn qubits

More Quantum Algorithms

Class	Problem/Algorithm	Paradigm Used
Inverse Function	Grover's Algorithm	GO
	Bernstein-Vazirani	n.a
Number-theoretic Applications	Shor's Factoring Algorithm	QFT
Algebraic Applications	Linear Systems	HHL
	Matrix Element Group Representations	QFT
	Matrix Product Verification	GO
	Subgroup Isomorphism	QFT
	Persistent Homology	GO,QFT
Graph Applications	Graph Properties Verification	GO
	Minimum Spanning Tree	GO
	Maximum Flow	GO
	Approximate Quantum Algorithms	SIM
Learning Applications	Quantum Principal Component Analysis (PCA)	QFT
	Quantum Support Vector Machines (SVM)	QFT
	Partition Function	QFT
Quantum Simulation	Schrodinger Equation Simulation	SIM
	Transverse Ising Model Simulation	VQE
Quantum Utilities	State Preparation	n.a
	Quantum Tomography	n.a.
	Quantum Error Correction	n.a.

Quantum Algorithm Zoo:
https://quantumalgorithmzoo.org/

Quantum Programming

- There is already a number of programming languages adapted for quantum computing
- The purpose of quantum programming languages is to provide a tool for researchers, not a tool for programmers
- **QCL** is an example of such language
- **IBM QISKit** (Quantum Information Science Kit) is another example

Quantum Programming

QCL (**Q**uantum **C**omputation **L**anguage)

```
/* Remove "//" if starting interpreter with -n option */
// extern operator H(qureg q);

procedure FlipCoin() {
  qureg q[1]; int x;

  reset;
  H(q);
  measure q, x;
  if x == 1  { print "Heads"; }
  if x == 0  { print "Tails"; }
  reset;
}
```

C-like syntax

allows combining of quantum and classical code

http://tph.tuwien.ac.at/~oemer/qcl.html

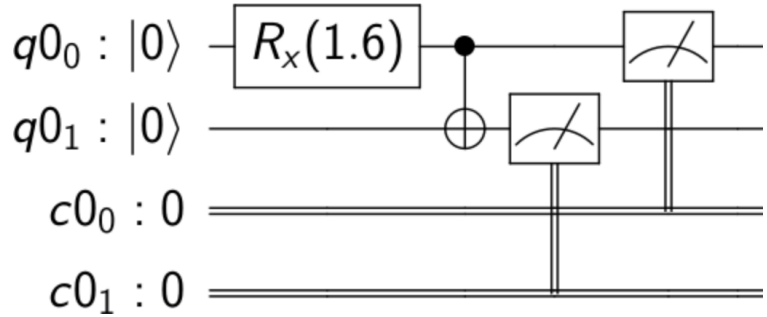
Quantum Programming

QISKit (Quantum Information Science Kit)

- **Qiskit** is an open-source framework for quantum computing.
- **It provides tools for creating and manipulating quantum programs** and running them on prototype quantum devices on IBM Quantum Experience over Cloud based access.

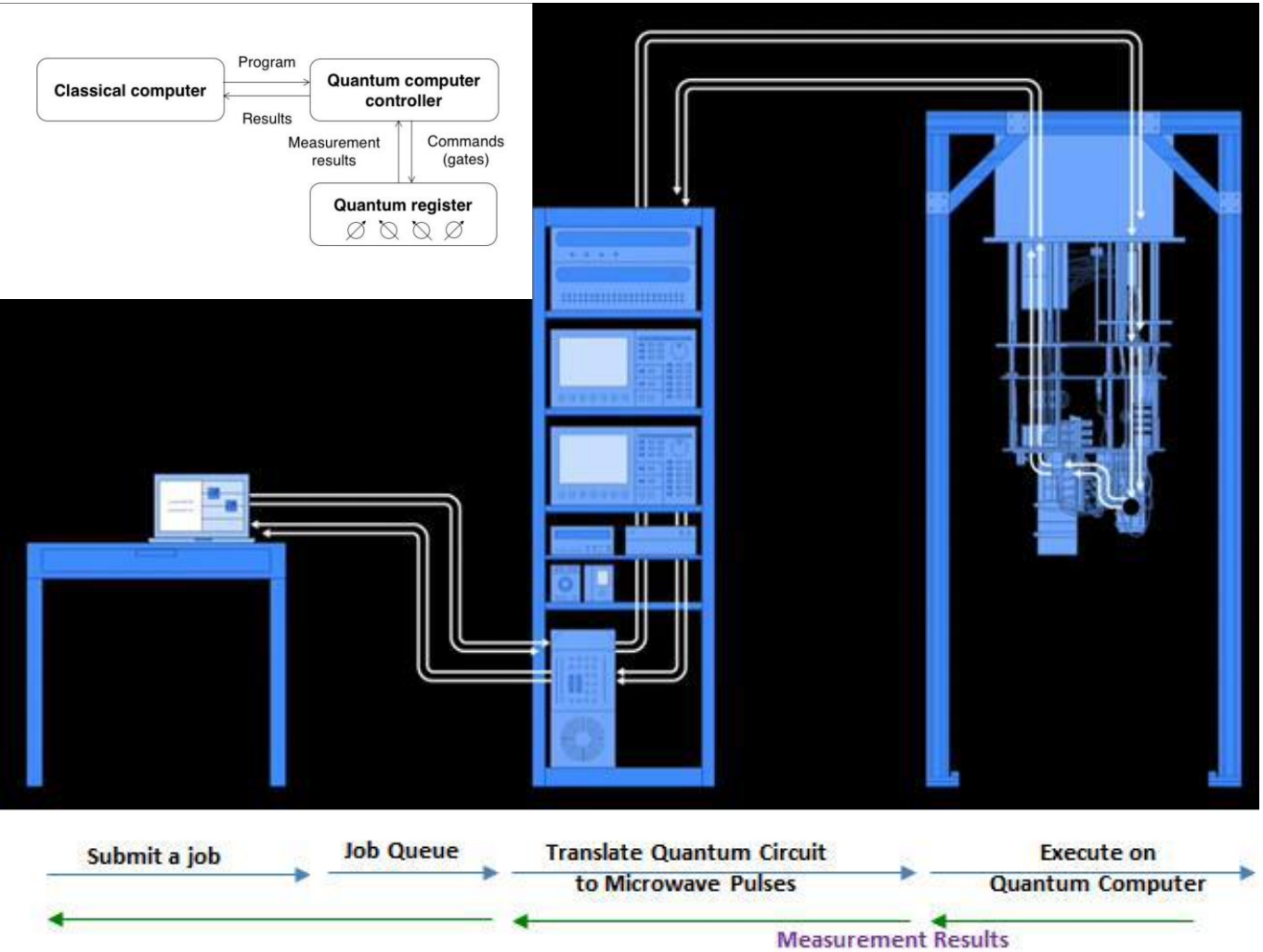
```
In [7]: from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
        from qiskit.tools.visualization import circuit_drawer
        import numpy as np

        qr = QuantumRegister(2)
        cr = ClassicalRegister(2)
        qp = QuantumCircuit(qr,cr)

        qp.rx( np.pi/2,qr[0])
        qp.cx(qr[0],qr[1])

        qp.measure(qr,cr)

        circuit_drawer(qp)
```

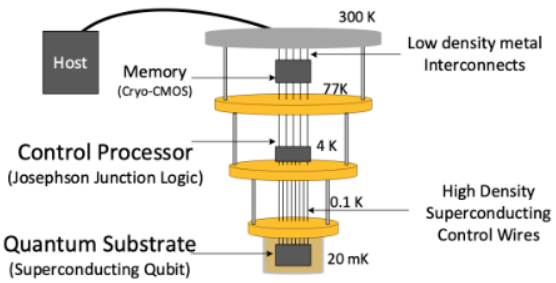
Out[7]:




```
In [1]: from qiskit import *

In [2]: qr = QuantumRegister(2)
        cr = ClassicalRegister(2)

In [3]: c = QuantumCircuit(qr, cr)    # c = QuantumCircuit(2,2)
```

How does a quantum computer work?

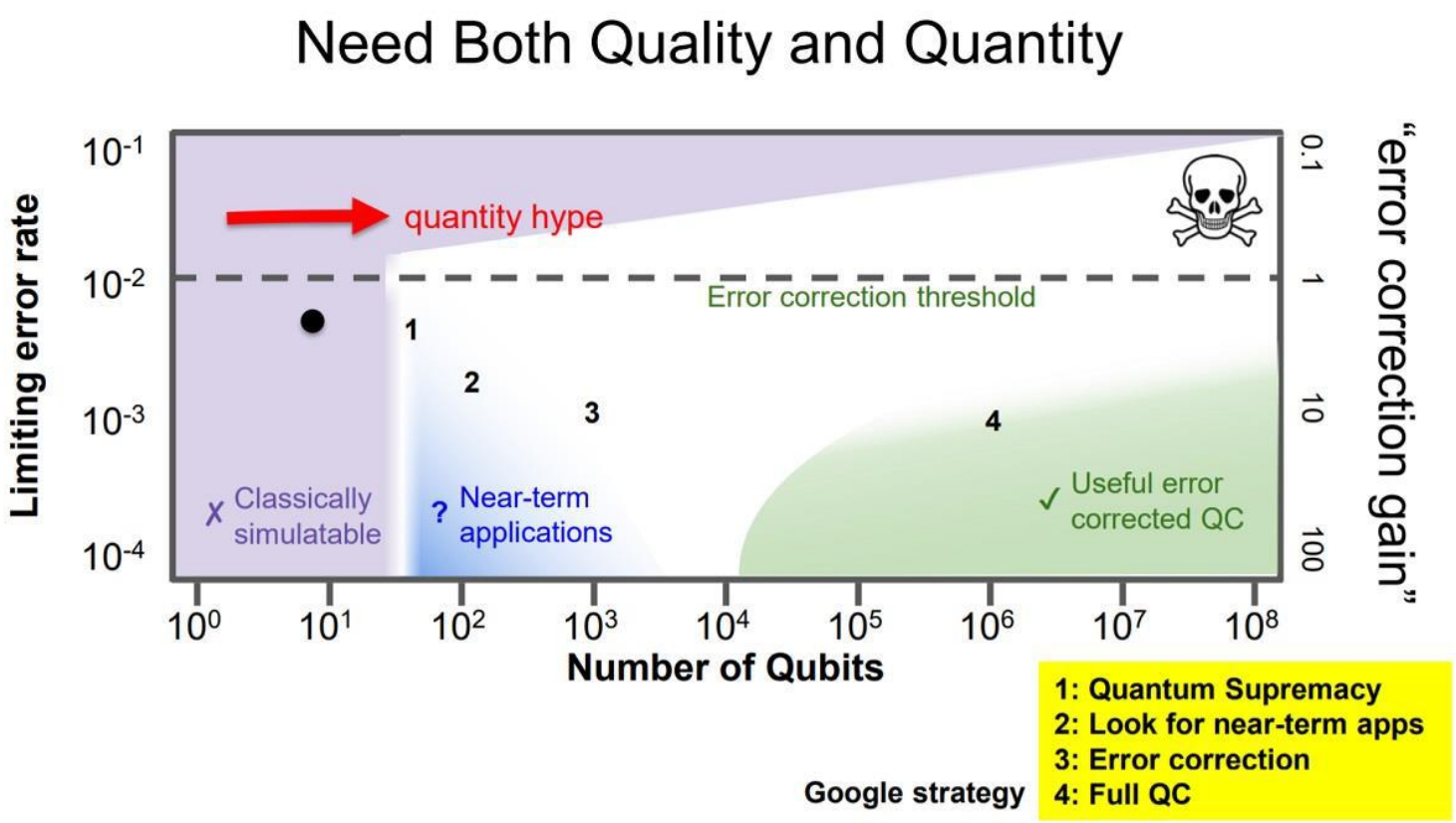




The flow of submitting a job from a classical computer to a quantum computer, executing the job, and returning quantum measurement results to the classical computer.

NISQ –Noisy Intermediate-Scale Quantum Era

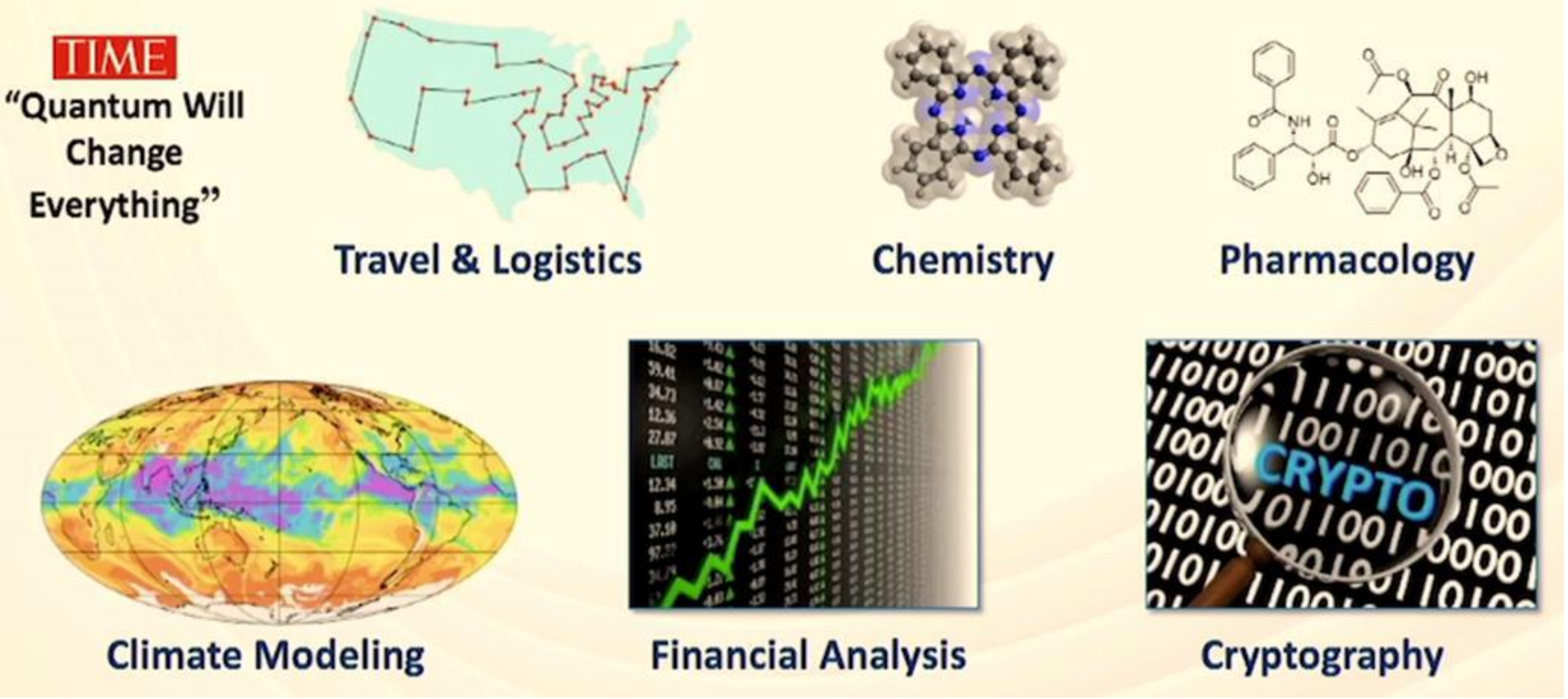
- John Preshill of CalTech introduced this terminology in 2018



Problems with Quantum Computers

- **Decoherence**
  - Quantum system is extremely sensitive to external environment, so it should be safely isolated
- It is hard to achieve the decoherence time that is more than the algorithm running time
- Error correction (requires more qubits!)
- Physical implementation of computations
- New quantum algorithms to solve more problems
- Entangled states for data transfer
- Running time increases exponentially with matrix size

Applications of Quantum Computers



Applications of Quantum Computers

Application 1. Quantum Factoring

P. Shor (1994)

A quantum computer can factor numbers **exponentially faster** than classical computers

15 = 3 × 5 (...easy)
3864788462100938762143232563 1 = ? × ?

Importance: cryptanalysis

public key cryptography relies on inability to factor large numbers

Best classical algorithm: 10^{24} steps	Shor's quantum algorithm: 10^{10} steps
On classical THz computer: 150,000 years	On quantum THz computer: <1 second

Application 2: Quantum Search

L. Grover (1997)

A quantum computer can find a marked entry in an unsorted database **quadratically faster** than classical computers

(e.g., given a phone number, finding the owner's name in a phonebook)

Importance: "satisfiability" problems

- fast searching of big data
- inverting complex functions
- determining the median or other global properties of data
- pattern recognition; machine vision

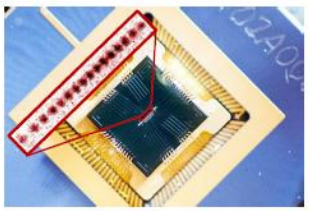
Applications of Quantum Computers

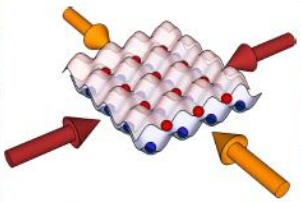
Application 3: Quantum Simulation

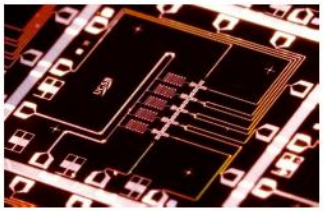
Quantum modelling is hard: *N* quantum systems require solution to 2^N coupled eqns

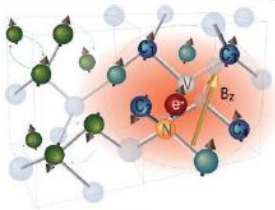
$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi$$

Alternative approach: Implement model of interacting system on a ***quantum simulator***, or "standard" set of qubits with programmable interactions

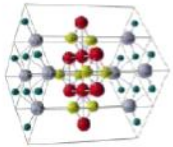

Atomic ions

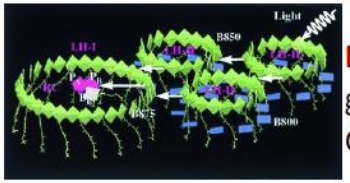

Trapped atoms


Superconductors


NV-diamond

Quantum Material Design Understand exotic material properties or design new quantum materials from the bottom up





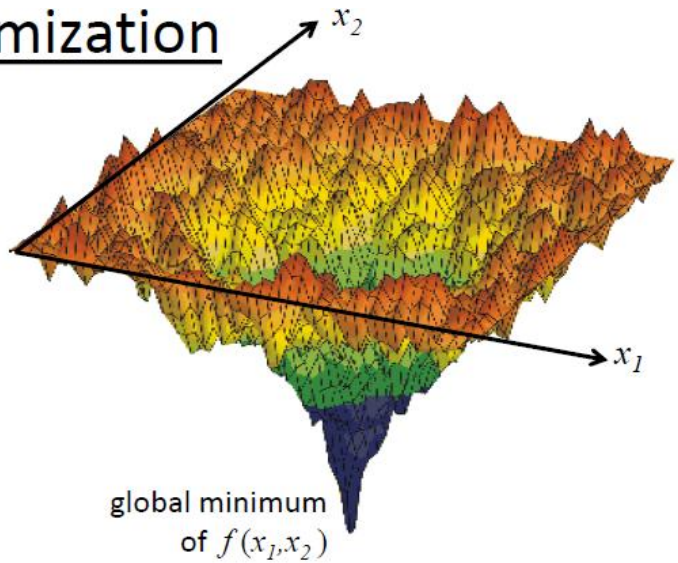
Energy and Light Harvesting Use quantum simulator to program QCD lattice gauge theories, test ideas connecting cosmology to information theory (AdS-CFT etc..)

Quantum Field Theories Program QCD lattice gauge theories, test ideas connecting cosmology to information theory (AdS-CFT etc..)

Applications of Quantum Computers

Application 4: Quantum Optimization

Minimizing complex (nonlinear) functions by "simultaneously sampling" entire space through quantum superposition



Relevant to

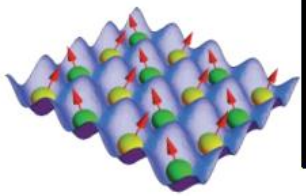
- Logistics
- Operations Research
- VLSI design
- Finance

Example: quadratic optimization

Minimize

$$f(x_1, x_2, \dots) = \sum_{i<j} q_{ij} x_i x_j + \sum_i c_i x_i$$

this function maps to energy of quantum magnetic network

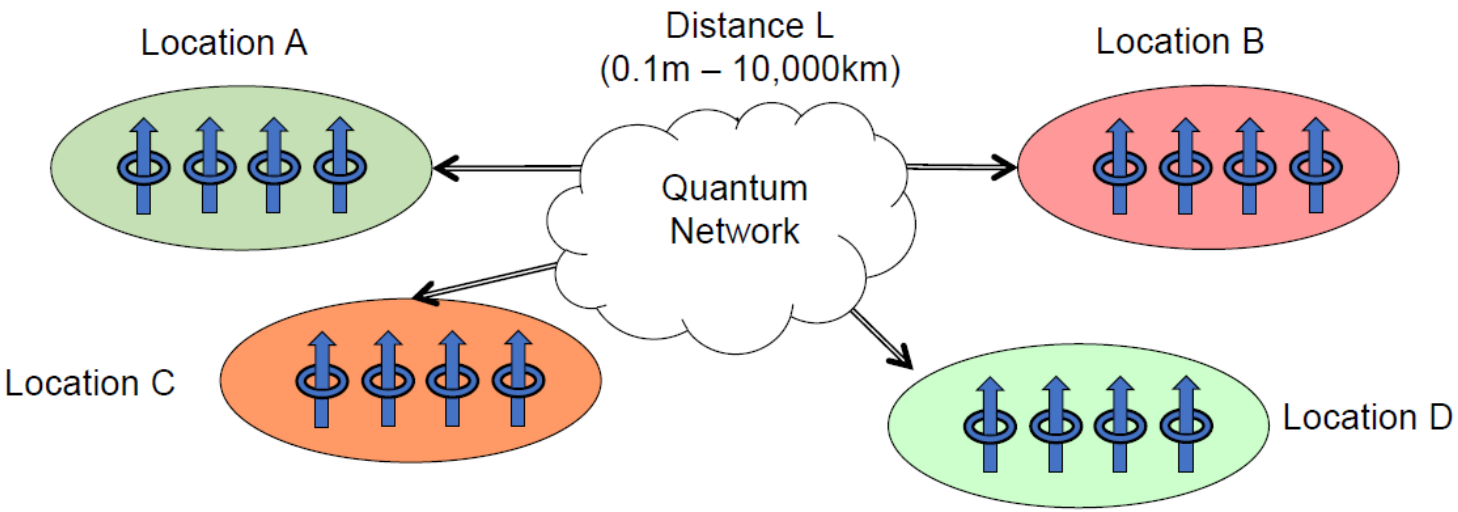


Killer Application?

- could crack a large class of intractable problems: factoring, "traveling salesman" problem, etc..
- BUT not known if there is always a quantum speedup

Applications of Quantum Computers

Application 5: Quantum Networks

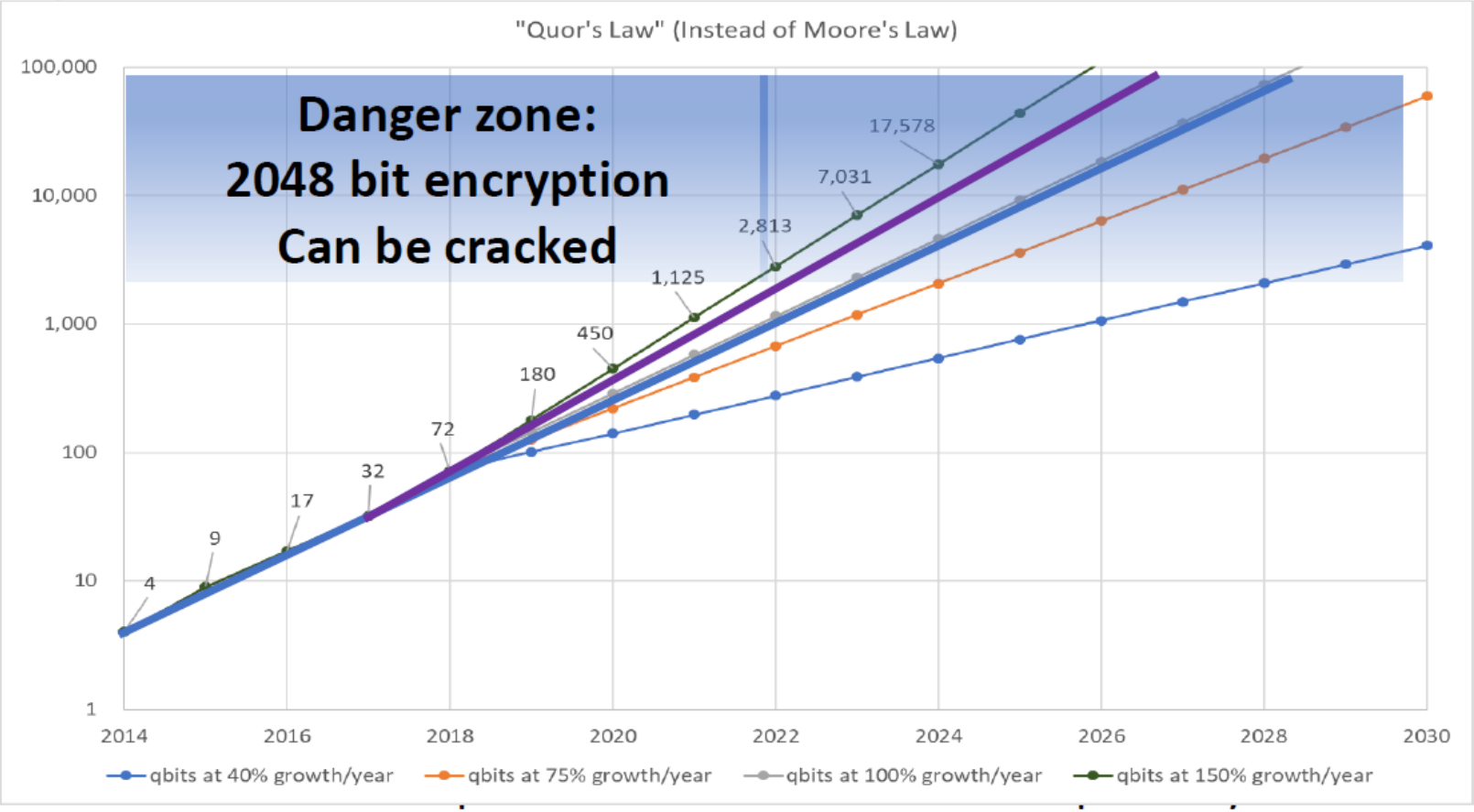


Uses of a quantum network

- Secret key generation: cryptography
- Certifiable random number generation
- Quantum repeaters ("amplifiers")
- Distributed quantum entanglement for optimal decision making
- *Large-scale quantum computing*

Post-Quantum World

- **There is a good chance that quantum computers will be able to crack RSA-2048 within five-ten years (need about 8,000 qubits in a universal computer to do this).**
  - Some encrypted data has a shelf-life of more than ten years. It may take ten years to cut over to a new encryption scheme, so companies and governments are scrambling to figure out what to do.
- **Most encryption used today is not safe in a quantum world.**
  - If someone has been recording a https session, say, they may not be able to decrypt it now, but a few years from now, who knows.
- **There are encryption algorithms that are safe:**
  - Secure communications will move to lattice algorithms.
- **QKD (Quantum Key Distribution) will likely be more popular for the most sensitive communications**
  - Some governments and banks are already starting to use “Quantum Safe” algorithms, and this is expected to increase over the next decade.

Future of Quantum Computers

Qubit count of general-purpose quantum computers will double roughly every one to two years, or roughly every 18 months on average.